SCHOOL MANAGEMENT SYSTEM FOR KYAMATU PRIMARY SCHOOL

FINAL PROJECT REPORT

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ABSTRACT

This report presents the design, development, and deployment of a web-based School Management System for Kyamatu Primary School, Kitui County. The system automates student management, attendance tracking, CBC-aligned assessment and grading, report card generation, fee management, timetabling, and communication. Built with React 18, Node.js/Express, and PostgreSQL, the system was deployed on Cloudflare Pages and Render. All 11 modules were successfully implemented, tested, and deployed, achieving 100% functional requirement coverage.

CHAPTER 1: INTRODUCTION

(See Proposal Document for complete Background, Problem Statement, Objectives, and Significance sections.)

CHAPTER 2: LITERATURE REVIEW

(See Proposal Document for complete Literature Review.)

CHAPTER 3: METHODOLOGY

(See Proposal Document for complete Research and Development Methodology.)

CHAPTER 4: SYSTEM DESIGN

(See Design Specification Document for complete System Architecture, Database Design, Module Design, Frontend Design, and Security Design.)

CHAPTER 5: IMPLEMENTATION

5.1 System Completion Status

Module	Status	Completion
Authentication & RBAC	Complete	100%
Student Management	Complete	100%
Staff Management	Complete	100%
Academic Structure (CBC)	Complete	100%
Attendance Tracking	Complete	100%
Assessment & Grading	Complete	100%
Report Cards	Complete	100%
Fees Management	Complete	100%
Communication	Complete	100%
Timetable	Complete	100%
Dashboard (Admin + Student)	Complete	100%
Deployment	Complete	100%

5.2 Database Statistics

23+ database tables, 6 grades, 7 classes, 10 teachers, 70+ students, full assessment data with scores, fee structures with invoices and payments, auto-generated timetable.

5.3 Key Technical Achievements

- Fully automated CBC competency grading engine
- Role-based dashboards with real-time analytics
- Audit logging for all critical system operations
- Redis-backed rate limiting for API security
- Auto-repair mechanism for timetable and assessment data on server startup

CHAPTER 6: TESTING

(See Test Plan Document for complete test strategy, test cases, and results.)

Summary: 46 test cases executed across all modules. 100% pass rate achieved.

CHAPTER 7: CONCLUSION AND RECOMMENDATIONS

7.1 Conclusion

The School Management System for Kyamatu Primary School has been successfully designed, developed, tested, and deployed. The system addresses all identified problems: manual record-keeping, data loss risk, slow report generation, poor parent communication, and lack of real-time analytics. All 11 core modules are fully functional and accessible to stakeholders.

7.2 Recommendations for Future Work

- M-Pesa STK Push Integration: Direct mobile money payment initiation from within the system.
- SMS/Email Notifications: Automated alerts for attendance, fees, and announcements.
- Flutter Parent App: Native mobile application for parents to access student data.
- PDF Report Card Generation: Server-side PDF generation for downloadable report cards.
- Multi-School Support: Architecture extension to support multiple schools under one platform.

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